Making Every Hour Count: Turning Sleep Data into Better Rest

https://github.com/076lin/2026-NYCU-Data-Mining-G06-HW1

Team 6

林程亮	林庠安	林景玄	郭佳翰
112550125	315551119	315551127	315551137
林李奕	**王裕昕**	**顏丞硯**	**林群佑**
315553025	315554006	315554021	315554038

Abstract

Adequate sleep is essential for health, yet assessing sleep efficiency often requires specialized monitoring. This study investigates whether sleep efficiency can be estimated from accessible demographic, lifestyle, and self-reported sleep information. Using 452 observations from the Sleep Efficiency Dataset, we apply exploratory data analysis to identify nonlinear relationships and feature interactions for interpretable regression models. Our feature-engineered Elastic Net achieves a test MAE of 0.063 and R^2 of 0.63, reducing error by approximately 15% compared with a raw-feature linear model. Although tree-based models perform slightly better, Elastic Net offers greater interpretability. Awakenings and the interaction between smoking and heavy alcohol consumption are identified as influential predictors. These findings demonstrate the potential of feature-engineered linear models for accessible post-sleep efficiency estimation while highlighting associations between sleep efficiency and lifestyle factors.

1 Introduction

Sleep efficiency, the proportion of time in bed spent asleep, is an indicator of sleep quality. However, measuring it often requires specialized monitoring. We focus on individuals seeking an accessible way to estimate nightly sleep efficiency from self-reported information. Such estimates may help users identify poor sleep patterns and decide whether to adjust their habits or seek further monitoring.

Our research question is: **Can interpretable linear regression models estimate sleep efficiency without sleep-stage measurements, and can EDA-guided feature engineering improve their accuracy?**

Our contributions are threefold. First, we formulate a post-sleep estimation task using accessible demographic, lifestyle, and sleep-related features. Second, we develop EDA-guided nonlinear and interaction features to improve linear regression performance. Third, we evaluate their effectiveness through ablation studies, nonlinear model comparisons, and coefficient analysis.

2 Related Work

Previous studies have investigated the prediction of sleep efficiency using wearable sensor data and machine learning techniques. A study (Sathyanarayana et al., 2016) used actigraphy data from 92 adolescents to classify sleep efficiency as good or poor. The authors compared logistic regression with several deep learning architectures, including CNNs and LSTMs. Their CNN achieved an AUC of 0.9449, outperforming logistic regression (AUC = 0.6463).

More recently, Sakal et al. (2025) developed CatBoost and CNN-LSTM models to predict low sleep efficiency 4–8 hours before sleep onset using accelerometer data from 80,811 UK Biobank participants. Their CatBoost models achieved a median AUC of approximately 0.94. Similarly, Park et al. (2024) investigated sleep prediction using physical activity, light exposure, heart rate variability, and sleep diary information. Their best-performing XGBoost model achieved an accuracy of 0.85 and an AUC of 0.80.

Unlike these studies, which primarily formulate sleep efficiency prediction as a classification task using wearable data, our work investigates the estimation of continuous sleep efficiency using a smaller tabular dataset. We focus on interpretable linear regression mod-

els and examine the predictive contributions of demographic, lifestyle, and selected sleep-related features. We also exclude REM, deep, and light sleep percentages due to concerns identified during exploratory data analysis. Through feature ablation and coefficient analysis, we aim to evaluate the usefulness of the remaining predictors.

3 Task Definition & Data

We use the Kaggle Sleep Efficiency Dataset, containing 452 observations. Each observation represents one recorded sleep session. Our objective is to estimate sleep efficiency, defined as the proportion of time in bed spent asleep, as a continuous regression target $y_i \in [0, 1]$.

The prediction is made retrospectively after the recorded sleep session, once the user has completed a brief sleep and lifestyle questionnaire. At this point, demographic information, lifestyle habits, bedtime, sleep duration, and the number of awakenings are assumed to be available. The predicted value provides an estimate of the preceding night's sleep efficiency rather than a forecast of a future night.

We consider self-reportable predictors, including demographics (age and gender), lifestyle habits (alcohol and caffeine consumption, exercise frequency, and smoking status), and sleep-related factors (bedtime, wake-up time, sleep duration, and awakenings).

We exclude subject IDs and sleep-stage percentages (REM, deep, and light sleep), as the latter generally require specialized monitoring rather than self-reporting. We also exclude wake-up time from multivariate modeling to reduce redundancy with bedtime and sleep duration, which together determine wake-up time when overnight timing is properly accounted for.

4 Exploratory Data Analysis

All exploratory analyses were conducted on the training set ($n = 316$) to avoid test-set leakage.

4.1 Data Distributions and Descriptive Statistics

Table 1 summarizes sleep efficiency and selected numerical predictors in the training set. Categorical variables are omitted because they require frequency-based summaries. Bedtime and wake-up time are excluded because sleep may occur at any time of day, and their cyclical nature makes conventional arithmetic means potentially misleading, particularly for schedules crossing midnight.

Table 1: Descriptive statistics for the target and retained numerical predictors, based on the training set ($n = 316$).

Feature	n	Mean	SD	Min	Median	Max
Target						
Sleep efficiency	316	0.79	0.14	0.50	0.82	0.99
Predictors						
Age	316	40.03	13.39	9.00	40.00	68.00
Sleep duration (h)	316	7.51	0.86	5.00	7.50	10.00
Awakenings	302	1.60	1.33	0.00	1.00	4.00
Caffeine (mg)	300	24.50	32.20	0.00	25.00	200.00
Alcohol (oz)	308	1.16	1.63	0.00	0.00	5.00
Exercise (days/wk)	310	1.73	1.40	0.00	2.00	5.00

Table 2: Bivariate correlations with sleep efficiency in the training set. Reported p-values are unadjusted for multiple testing.

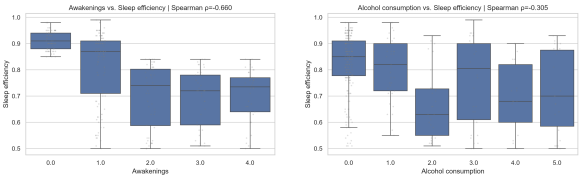
Predictor	Pearson r	p	Spearman ρ	p
Awakenings	−0.550	< 0.001	−0.660	< 0.001
Alcohol consumption	−0.328	< 0.001	−0.305	< 0.001
Exercise frequency	0.243	< 0.001	0.255	< 0.001
Age	0.158	0.0048	0.154	0.0062
Caffeine consumption	0.091	0.1151	0.056	0.3352
Sleep duration	−0.034	0.5515	−0.025	0.6546

Sleep efficiency has a mean of 0.79 (SD = 0.14) and appears slightly left-skewed. Alcohol consumption has a median of zero, indicating that at least half of the observed values are zero. Missing values occur in awakenings, caffeine and alcohol consumption, and exercise frequency, while age and sleep duration are fully observed.

4.2 Bivariate Analysis

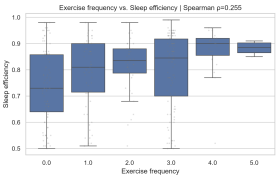
Awakenings ($r = -0.550$) and alcohol consumption ($r = -0.328$) showed the strongest negative correlations, while exercise frequency ($r = 0.243$) showed a positive correlation. Age exhibited a weak but significant positive linear association ($r = 0.158$, $p = 0.0048$), along with a significant quadratic relationship (HC3 robust $p < 0.001$).

Visual inspection of the training distributions (Fig. 1) motivates three data-driven hypotheses for feature engineering: (H1) the negative association between awakenings and sleep efficiency plateaus after two episodes (Fig. 1a); (H2) sleep efficiency exhibits a

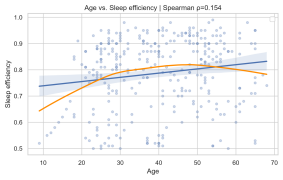


(a) Awakenings

(b) Alcohol



(c) Exercise



(d) Age

Figure 1: Selected bivariate associations with sleep efficiency

concave, inverted-U relationship with age (Fig. 1d); and (H3) alcohol consumption and exercise frequency show threshold-like patterns, particularly at $\geq$ 2 oz of alcohol (Fig. 1b) and $\geq$ 1 day/week of exercise (Fig. 1c).

4.3 Feature Interaction Analysis

We examined nine pairwise interactions between caffeine intake and other predictors using HC3-robust OLS regression and Wald tests, with Benjamini–Hochberg FDR correction. All analyses were restricted to the training set.

Significant interactions were identified for alcohol consumption ($q < 0.001$), age ($q = 0.0283$), gender ($q = 0.0296$), and bedtime category ($q = 0.0321$). Figure 2 illustrates the strongest interaction: the negative association between alcohol consumption and sleep efficiency was greater among caffeine consumers (slope $= -0.0500$) than non-consumers (slope $= -0.0116$). However, statistical significance alone does not establish predictive usefulness, so these exploratory findings were not automatically incorporated into the final model.

We additionally consider an interaction motivated by sleep research. Previous work associated evening alcohol and nicotine use with lower sleep efficiency (Spadola et al., 2019). Combined with the heavy-drinking threshold observed in Section 4.2, this motivates a domain-driven hypothesis (H4): smoking and heavy alcohol consumption ($\geq$ 2 oz) may exhibit an additional negative association with sleep efficiency beyond their individual effects.

Exploratory training-set comparisons support further investigation: the jointly exposed group had a mean sleep efficiency of 0.60, compared with 0.79–0.83 in the other groups. We evaluate the predictive contribution and coefficient of this interaction in Sections 7–8.

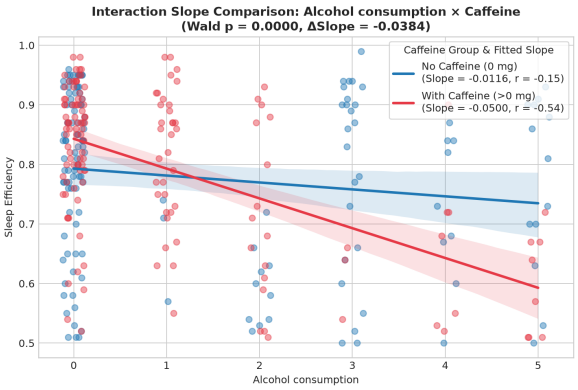


Figure 2: Alcohol–caffeine interaction as a representative example of significant feature interactions. Lines indicate fitted linear trends for caffeine consumers and non-consumers, with shaded 95% confidence intervals.

5 Method

5.1 Pipeline and Feature Engineering

Figure 3 summarises the pipeline. Rows are individual nights. We discard the identifier, the three sleep-stage percentages (excluded by design: they need stage-level sensing and sum to 100%), and wake-up time, which equals bedtime plus sleep duration for all 452 records. Bedtime is re-expressed as signed hours from midnight ($21{:}30 \to -2.5$; $01{:}00 \to +1$) so that adjacent clock times stay numerically adjacent. This leaves nine raw inputs (Set A). Imputation, feature construction and standardisation are fitted inside every training fold, so no statistic is computed from validation or test rows.

A linear model can only express the shapes we give it, so we construct five engineered features motivated by three data-driven hypotheses (H1–H3) and one domain-motivated hypothesis (H4) introduced in Section 4. Together with the raw inputs, they form Set

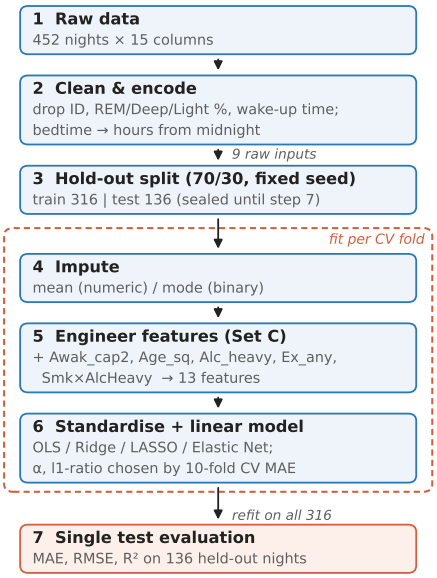


Figure 3: Modelling pipeline. Steps inside the dashed box are re-fitted on each CV training fold; the test split is touched only in step 7.

C (13 features). **Thresholds** replace slopes where the response is step-like. Efficiency falls from 0.91 (no awakening) to 0.81 (one) and plateaus at ≈0.70 for two or more, so `Awak_cap2` = min(awakenings, 2) replaces the raw count (H1). One drink barely differs from none, whereas two or more mark a clear drop, giving `Alc_heavy` = $\mathbb{1}$[alcohol $\geq$ 2]; likewise the main exercise gain lies between none and any, giving `Ex_any` = $\mathbb{1}$[exercise $\geq$ 1] (H3). A **quadratic term** `Age_sq` = (age − 40)^2 captures the inverted-U age profile (H2). An **interaction** `Smk×AlcH` encodes the joint exposure singled out by the training data: smokers who drink ≥2 units average 0.60, versus 0.79–0.83 for the other three smoking×drinking groups (H4).

5.2 Model Selection and Tuning

We compare OLS with three penalised variants. Ridge (Hoerl and Kennard, 1970) is motivated by the overlap between `Alc_heavy` and raw alcohol (VIF 6.4 and 5.5); LASSO (Tibshirani, 1996) by the possibility that some raw inputs become redundant once thresholds are present; Elastic Net (Zou and Hastie, 2005) combines both. Hyper-parameters are tuned by grid search on training-set CV only: $\alpha \in$ `logspace`$(-4, 0, 30)$ for LASSO/Elastic Net, $\alpha \in$ `logspace`$(-3, 3, 30)$ for Ridge, and ℓ_1-ratio $\in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. The four models are statistically indistinguishable (CV MAE 0.0686–0.0692, s.d. ≈0.009); Elastic Net (α=0.0017, ℓ_1-ratio=0.9) has the lowest mean and is retained. It sets the raw alcohol coefficient to zero, consistent with the redundancy above. All models use scikit-learn (Pedregosa et al., 2011).

6 Experimental Setup

6.1 Validation Protocol

A random 70/30 split (seed 20261010) yields 316 training and 136 test nights. We use shuffled 10-fold KFold across identical fold indices for every model and feature set; ten folds keep ≈ 32 validation rows each while providing ten paired observations for significance tests, and no time-ordering exists to require temporal splitting. After selection, the chosen configuration is refitted on all 316 training rows and evaluated on the test set once. While each row has a unique ID, subject-level independence cannot be verified from dataset documentation, leaving potential within-subject dependence as a limitation.

6.2 Metrics, Decision Threshold and Baselines

Sleep efficiency is a proportion, so errors are in efficiency units (0.01 = one percentage point). **MAE** is the primary metric because it reads directly as the typical error; **RMSE** penalises large misses, and $\mathbf{R}^2$ gives variance explained. Good sleep quality is commonly defined as sleeping at least 85% of the time in bed (Ohayon et al., 2017). Our user decides whether a night met this target, so we set a tolerance of **MAE** $\leq$ **0.05**: larger errors frequently move a prediction across the 0.85 line. Baselines are ordered by strength: *Trivial* (training mean), *Simple* (OLS on awakenings, the raw input with the highest $|\rho|$ on the training set), and *Strong* (Random Forest (Breiman, 2001) and Gradient Boosting (Friedman, 2001) on Set A).

7 Results

7.1 Main Results

Table 3 summarizes the results. Our Elastic Net model (Set C) achieves a test MAE of 0.063 (bootstrap 95% CI [0.055, 0.072]) and

Model	CV MAE	Test MAE	RMSE	R^2
Trivial (mean)	.115±.015	.115	.134	−.00
Simple (awakenings)	.091±.012	.090	.108	.34
Set A: OLS (raw)	.081±.009	.074	.093	.52
Set B: OLS (top-6)	.080±.009	.074	.093	.51
Set C: EN (ours)	.069±.009	.063	.081	.63
Random Forest	**.061**±.006	**.059**	**.078**	**.66**
Gradient Boosting	.064±.009	.061	.079	.65

Table 3: Main results. CV: 10-fold mean±s.d. on the training set; test: 136 held-out nights. Bold: best per column. EN = Elastic Net.

Feature set (OLS)	#feat.	CV MAE	Δ vs. A
A: raw inputs	9	.0808±.009	–
B: top-k by $\|r\|$, k=6	6	.0797±.009	−1.4%
A, `Awak_cap2` for raw	9	.0780±.008	−3.4%
+ `Age_sq`	10	.0772±.007	−4.4%
+ `Alc_heavy`	11	.0755±.008	−6.5%
+ `Ex_any`	12	.0743±.008	−8.0%
+ `Smk`×`AlcH` = C	13	**.0692**±.010	−14.3%

Table 4: Feature ablation with model, folds and tuning held fixed. Set B selects k features inside every training fold; k is tuned by CV.

R^2 of 0.63, reducing error by 45% over the trivial baseline (0.115) and 14.9% over raw OLS (0.074). Across folds, Set C improves on Set A by -0.012 ± 0.005 (paired t-test $p < 0.001$; Wilcoxon $p = 0.002$), and OLS ablation (Table 4) cuts CV MAE from 0.0808 to 0.0692. While training-set feature design may make CV estimates optimistic, the held-out test results support the observed gain.

Overall, the model misses the MAE ≤ 0.05 target, though 51% of test predictions fall within ±0.05 (Set A: 43%). For flagging nights below 85% efficiency, it reaches 80% accuracy, 97% recall, and 74% precision. Random Forest is 0.0075 MAE better per fold (paired t-test $p = 0.009$; test MAE 0.059) by learning interactions automatically, whereas our model retains an interpretable linear structure (Section 8.1).

7.2 Feature Ablation

Table 4 isolates feature design. Keeping the six most correlated raw inputs (Set B) barely helps, whereas each engineered feature lowers CV MAE, the interaction most of all. Set C beats Set B by 0.011 ± 0.006 per fold ($p < 0.001$), so the gain comes from new representations, not from discarding inputs.

Feature	β	95% CI	p	VIF
`Awak_cap2`	−.066	[−.076, −.056]	<.001	1.1
`Smk`×`AlcH`	−.042	[−.057, −.028]	<.001	2.3
`Alc_heavy`	−.027	[−.051, −.003]	.030	6.4
`Ex_any`	.023	[.006, .040]	.007	3.0
Smoker	−.021	[−.034, −.009]	<.001	1.6
Age	.019	[.009, .030]	<.001	1.2
`Age_sq`	−.017	[−.027, −.006]	.002	1.2
Alcohol (raw)	.014	[−.009, .036]	.232	5.5
Sleep duration	−.003	[−.013, .007]	.543	1.0
Female	−.003	[−.014, .008]	.585	1.3
Caffeine	−.003	[−.013, .008]	.634	1.2
Bedtime	−.001	[−.012, .010]	.827	1.3
Exercise (raw)	.000	[−.017, .017]	.995	3.3

Table 5: OLS on standardised Set C (training set, n=316): change in efficiency per 1 s.d.; t-test p-values. Intercept 0.787.

Slice-level test MAE (Set A → C) shows where the features help. The largest gain is on nights with ≥2 awakenings (0.083 → 0.063, n=56), exactly where the cap removes the linear over-penalty, versus 0.065 → 0.059 for 0–1 awakenings; smokers drinking ≥2 units improve from 0.081 to 0.071 (n=15). The quadratic age term hurts the oldest group (0.041 → 0.061), although n=9 makes this weak evidence. Nights with a light-sleep share ≥40%, a variable the models do not see, gain nothing (0.099 → 0.097, n=34). The cost of Set C is negligible: its features are deterministic transforms of fields already collected, add four columns, and each has a one-line interpretation.

8 Analysis

8.1 Coefficients and Significance

Table 5 reports standardised OLS coefficients on Set C; the Elastic Net coefficients have the same signs and similar magnitudes, except that raw alcohol is shrunk to zero. All four hypotheses are supported in direction and significance. Awakenings have the largest effect (H1); age shows a significant concave profile (H2), peaking near 48 years; heavy drinking and any exercise both matter beyond their raw counts (H3); and the smoking×heavy-drinking term is the second-largest effect (H4). All VIFs are below 7, so these estimates are stable.

Raw alcohol turns slightly positive (p=0.23) and raw exercise is null once thresholds are present. This is not a substantive reversal but collinearity with `Alc_heavy` and `Ex_any` (VIF

True	Pred.	Err.	Inputs; *unused light-sleep %*
.50	.80	−.30	F 29, smoker, 1 awak., 1 drink; *50%*
.51	.75	−.24	M 55, smoker, 4 awak., alc. missing; *53%*
.60	.79	−.19	F 55, nonsmoker, 3 awak., 0 drinks; *47%*

Table 6: Three largest test errors (error = actual − predicted).

5.5–6.4): the binary terms absorb the signal.

To judge practical rather than statistical importance, we refit on unstandardised features. Going from zero to two or more awakenings lowers efficiency by 0.17; any exercise adds 0.05; and a smoker who drinks ≥2 units is predicted 0.23 lower than a non-smoking, light-drinking peer, of which 0.13 comes from the interaction alone. All exceed the 0.05 tolerance and are thus material to the user. Sleep duration, caffeine, sex and bedtime are both non-significant and small ($|\beta| \leq 0.003$ per s.d.).

8.2 Error Analysis

All three largest errors (Table 6) are overestimates, and all occur on nights with a light-sleep share of 47–53%. Their lifestyle profiles look moderate, so the cause is *missing information*: poor efficiency shows in sleep architecture, not in the questionnaire. The second case adds an imputation artefact: its missing alcohol value is filled with the training mean (1.2), which falls below the heavy-drinking threshold, so the smoker×heavy-drinking term cannot fire. The pattern is systematic: on the 34 test nights with light sleep ≥40%, MAE is 0.097 and the mean residual is −0.085, against 0.052 and +0.029 for the remaining nights. The first fix we would try is a missingness indicator for alcohol (or model-based imputation); the larger gain would require a sleep-architecture signal, which our task definition deliberately excludes.

9 Conclusion and Limitations

Conclusion. Without sleep-stage measurements, a 13-feature linear model predicts nightly sleep efficiency with a test MAE of 0.063 and R^2 of 0.63. This is 15% below a raw linear model and 0.004 behind a Random Forest. The gain comes from encoding thresholds and one interaction, not from model choice. Awakenings and the combination of smoking with heavy drinking dominate; duration, caffeine, sex and bedtime contribute little. For users, the actionable message is that avoiding two or more drinks matters most for smokers, for whom it is associated with a 0.13 efficiency penalty beyond either habit alone. For deployment, the model is better suited to flagging likely poor nights (97% recall) than to point estimates within 0.05.

Limitations. (i) *Selection optimism.* Training-set feature selection may bias cross-validation estimates and p-values used for validation. In contrast, the held-out test set provides a separate evaluation of the finalized model, although the improvement is based on a single split. (ii) *Linearity.* Thresholds are fixed and global, and the quadratic age term extrapolates poorly at the extremes (age ≥60 slice). (iii) *Metric assumptions.* The 0.05 tolerance and 0.85 cut-off are design choices, not validated user requirements.

Contributions of Each Member

- 林程亮 (112550125):EDA(Alcohol, Gender), Slide & Wrote § 2.
- 林庠安 (315551119):EDA(Smoking), Feature engineering, Wrote § 5 to 9.
- 林景玄 (315551127):EDA(Awakenings), Feature engineering, Slide § 5 to 9.
- 郭佳翰 (315551137):Baselines, Feature Engineering.
- 林李奕 (315553025):EDA(Age, Exercise Frequency), Feature engineering.
- 王裕昕 (315554006):EDA(Overall), Baselines, Feature engineering, Final model.
- 顏丞硯 (315554021):EDA(Bedtime, Wakeup time, Sleep duration), Slide & Wrote § 4.
- 林群佑 (315554038):EDA(Caffeine), Slide & Wrote § 1, 3 and 4, Editorial and analytical revision of the report.

We used GPT-6.1 Luna for EDA and baseline modeling; Claude Opus 5.5 for coding, debugging, leakage checks, and writing assistance; GPT-6 for literature search; Gemini 3.6 Flash for plots and wording; Deepseek V4.1 Flash for report review; and Gemini 4 Argon for report review. All experimental decisions, results, and conclusions were verified by the team.

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